

This is a sample which very roughly reflects the structuring and style of questions in the exams.

You are supposed to answer any 2 of the 3 questions.

Each question has multiple parts (a,b,c), and carries a total of 50 marks.

Each question is on a mix of topics covered at different stages of the module, i.e., a mixture of electrostatics, magnetostatics and dynamics.

1. (a) An infinitely long charged wire has linear charge density λ . Use Gauss's dielectric flux theorem to calculate the electric field at a distance r from the wire. If you use any assumptions based on symmetry, state them clearly.

[20 pts.]

- (b) The current density in some region is described by

$$\mathbf{J} = \gamma \left(\frac{4y}{x^2 + y^2} \hat{i} - \frac{4x}{x^2 + y^2} \hat{j} \right)$$

How is the charge density ρ in this region changing? If you use an equation related to the conservation of charge, name this equation explicitly.

[15 pts.]

- (c) An electromagnetic system is described by the fields

$$\mathbf{E} = K_0 x \sin(\omega t) \hat{j} \quad \mathbf{B} = \frac{K_0}{\omega} \cos(\omega t) \hat{k}$$

Calculate the displacement current density $\mathbf{J}_D$.

Use Maxwell's equations to calculate the current density $\mathbf{J}$.

[15 pts.]

2. (a) An infinite wire carrying current I_1 runs along the y axis; the current flows from $y = -\infty$ to $y = +\infty$ through the origin. A square loop lies in the xy plane, with the four corners having coordinates (x_0, y_0) , $(x_0 + L, y_0)$, $(x_0 + L, y_0 + L)$, and $(x_0, y_0 + L)$. Current I_2 flows counterclockwise through the square loop.

Calculate the total force acting on the square loop due to the current in the long wire.

Hint: A sketch showing a top view of the xy plane, with arrows showing the directions of forces on the four sides, might help.

[20 pts.]

- (b) The electric field in some region is given by

$$\mathbf{E} = \frac{V_0 y}{x^2 + y^2} \hat{i} + \frac{V_0 x}{x^2 + y^2} \hat{j}$$

Calculate the charge density in this region.

[15 pts.]

- (c) A circular loop lies on a plane perpendicular to a uniform magnetic field of strength B_0 . The circular loop decreases its size over time: its radius is given by

$$r(t) = r_0 \left(1 - \frac{t}{t_0} \right)$$

Find the magnitude of the EMF induced in the loop.

[15 pts.]

3. (a) Ampere's theorem for the magnetic field due to a steady current is

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

What does the symbol I refer to?

Starting from this integral relation, derive part of Maxwell's fourth equation (which concerns the curl of the magnetic field). If you use a theorem about vector integrals on the way, name the theorem explicitly.

[20 pts.]

- (b) A solid metallic sphere has radius R , and carries a total charge Q . The metal is an excellent conductor.

In which part of the sphere is the charge located?

Write expressions for the electric field magnitude at a distance r from the center of the sphere, both inside the sphere ($r < R$) and outside the sphere ($r > R$). Write expressions for the electric potential as a function of r , both inside and outside the sphere.

Sketch a plot of the electric field magnitude as a function of the distance r from the center of the sphere. Your plot should run from $r = 0$ to $r = 2R$.

[15 pts.]

- (c) An electromagnetic field is described by the scalar potential V and vector potential $\mathbf{A}$, given by

$$V = -LB_0\omega z \cos(\omega t); \quad \mathbf{A} = B_0x \sin(\omega t)\hat{k}$$

where B_0 , L , and ω are positive constants. Find the electric and magnetic fields in this system. In which spatial position(s) does the electric field vanish?

[15 pts.]